

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Regular and Supplementary Summer 2024****Course: B. Tech.****Semester: VIII****Branch: Mechanical Engg./ Mechanical Engg. (Sandwich)/ Production Engg.****Subject Code: BTMEC802F****Subject Name: Non – Conventional Energy Resources****Max Marks: 60****Date: 15/06/2024****Duration: 3.00 Hr.****Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/ expected answer as per OBE or the course outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non – programmable scientific calculators is allowed.
4. Assume suitable data whenever necessary and mention it clearly.

		Level/CO	Marks
Q.1	Solve Any Two of the following.		
A)	Explain national energy strategies and national energy plan.		6
B)	What is fossil fuel? What are different alternatives for fossil fuel?		6
C)	List the advantages and disadvantages of non – conventional energy sources.		6
Q.2	Solve Any Two of the following.		
A)	Explain solar energy as alternative energy source.		6
B)	Explain with neat sketch solar flat plate collector a solar air heater.		6
C)	With neat sketch explain the working of solar pond.		6
Q.3	Solve Any Two of the following.		
A)	Explain briefly the parameters affecting the performance of flat plate collectors.		6
B)	Explain the following terms i) Declination angle ii) Surface azimuth angle iii) Hour angle iv) Zenith angle.		6
C)	Explain principle of working of a solar cell.		6
Q.4	Solve Any Two of the following.		
A)	What is principle of OTEC? Draw a neat labelled sketch of open cycle OTEC power plant.		6
B)	What are various types of rotors in wind mill? Draw a neat labelled sketch of horizontal axis wind mill turbine.		6
C)	Explain working principle of tidal power plant? Enlist types of tidal power plant. Also explain any one type of tidal power plant.		6
Q.5	Solve Any Two of the following.		
A)	What is the geothermal energy? Explain binary cycle geothermal power plant.		6
B)	Explain with neat sketch working of Lithium ion Batteries.		6
C)	Explain the working principle of magnetohydrodynamic (MHD) power generation.		6
